

Sanjeevani (AgriTrust) — Dataset Architecture Report

Integration Architecture for Sovereign Indian Agricultural & Remote Sensing Datasets | NABARD & DPI Aligned

1. Executive Summary & Objective

Sanjeevani ingests, cross-validates, and synthesizes **6 sovereign Government of India agricultural datasets** alongside Sentinel-2 satellite remote sensing to construct an explainable, zero-PII Credit Passport and Risk Assessment engine. This multi-source grounding enables formal financial institutions to underwrite smallholder farmers with high confidence, adhering to NABARD Scale of Finance norms, PMFBY climate loss benchmarks, and DPDP Act 2023 privacy requirements.

2. Public Agricultural Dataset Mapping Matrix

Dataset & Source	Primary Attributes	Ingestion Connector	ML Engine Target	Platform Value & Impact
1. AGMARKNET 2.0 <i>agmarknet.gov.in</i> (MoA&FW;)	Daily Mandi Arrivals, Modal Price, Min/Max Price, Commodity, Grade	agmarknet_ingestor.py	Model D ARIMA Price Predictor	Harvest revenue realization models; price volatility stress-testing for credit sizing.
2. National Horticulture Board (NHB) <i>nhb.gov.in</i>	State & District Acreage, Production Volume, Yield (MT/Ha), Multi-year trends	nhb_ingestor.py	Model B Repayment Engine	Calibrates district Scale of Finance for horticulture crops (Grapes, Pomegranate, Onion).
3. PM-KISAN Beneficiary <i>Data.gov.in</i> (Sep 2026)	Village & Gender-wise Beneficiary Count, DBT Installment disbursement status	pmkisan_ingestor.py	Identity & Land Verification	Validates landholder smallholder credentials; contributes ■ 6,000/yr guaranteed liquidity buffer.
4. PMFBY Crop Loss <i>pmfby.gov.in / Data.gov.in</i>	Historical Claims Ratio, Loss Cost Ratio (LCR), Insured Area, Crop Losses	pmfby_ingestor.py	Model C Crop & Region Risk	Trains Random Forest on hyper-local climate distress; determines default risk under drought.
5. Kisan Call Centre (KCC) <i>Data.gov.in / DAC&FW;</i>	Query Category (Pest/Weather/Seeds), District, Crop, Season, Advisory Notes	kcc_advisory_ingestor.py	NLP Early-Warning Engine	Identifies emerging regional pest surges; pushes localized advisories in Hindi, Tamil & Punjabi.
6. ICAR Disease Image Dataset <i>Data.gov.in / ICAR-IASRI</i>	Annotated Leaf/Crop Images, Pathology Class, Pest Severity, Growth Stage	icar_disease_ingestor.py	Computer Vision CNN (MobileNetV3)	Powers camera leaf scans on Farmer PWA; auto-applies yield haircut factor in credit limits.

3. Multi-Model Architecture Integration Flow

- A. Ingestion & Normalization:** Connectors asynchronously poll REST/CKAN endpoints on Data.gov.in, storing cleaned records into the Unified Feature Store with PostgreSQL caching.
- B. Remote Sensing Fusion:** Sentinel-2 10m NDVI vegetative indices are intersected with PMFBY district loss records and AGMARKNET mandi prices.
- C. Explainable Underwriting:** Model outputs populate the CreditEligibilityResponse schema with transparent factor breakdowns, ensuring full explainability for bank credit committees.

4. Sovereign Data Governance & Privacy Guarantee

- Zero-PII Transmission:** Lenders query credit scores and risk bounds using cryptographic Consent Tokens (DPDP Act 2023 compliant).

- **KCC & NABARD Rules:** Enforces Scale of Finance limits per crop-acre to eliminate over-indebtedness.
- **Real-Time Resynchronization:** AGMARKNET prices refresh daily at 18:00 IST; Satellite imagery updates on a 5-day cycle.